A Unified Microrobotic Visual-Perception Processor with 62.2-FPS/mm^2 and 103-uJ/Frame Navigation in 28nm

Shuyuan Zhang[1], Yujin Wang[1], Yifan He[1], Yuchen Su[3], Huazhong Yang[1,2], Yongpan Liu[1,2], Hongyang Jia[1,2]

[1]Department of Electronic Engineering, Tsinghua University, Beijing

[2]BNRist, Tsinghua University, Beijing

[3]Xingjian College, Tsinghua University, Beijing

Outline

- **Introduction**
- **Challenges of Unified Visual Perception Processor (VPP)**
- **Proposed VPP with Unified Inference and Localization**
  - Architectural overview
  - Reconfigurable PE array for efficient fusion of MVM and linear solver
  - Unified dataflow with VLIW-style ISA for local buffer reduction
  - Flexible data mapping with a multi-granularity operand buffer
- **Measurement Results**
- **Conclusion**

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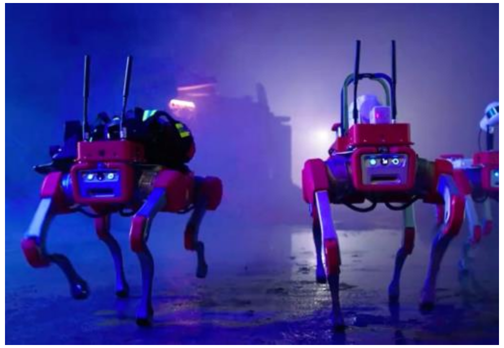
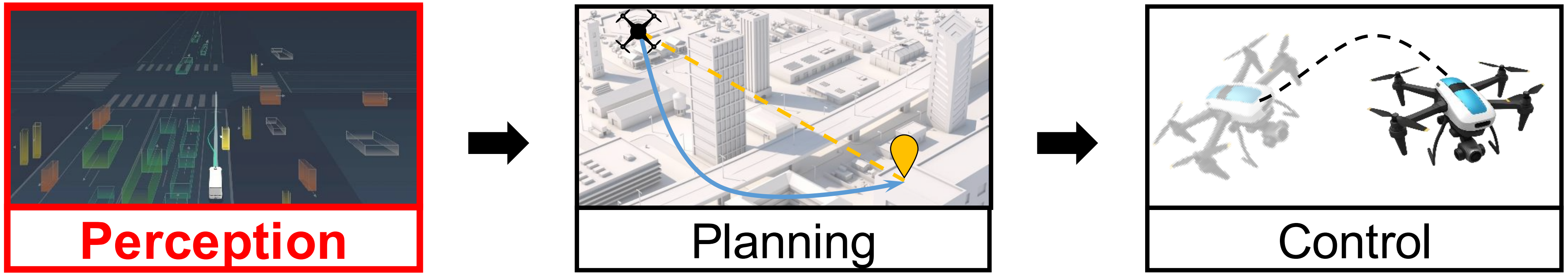
Autonomous robots

- **Autonomous robots in aerial photography, package delivery and rescue…**

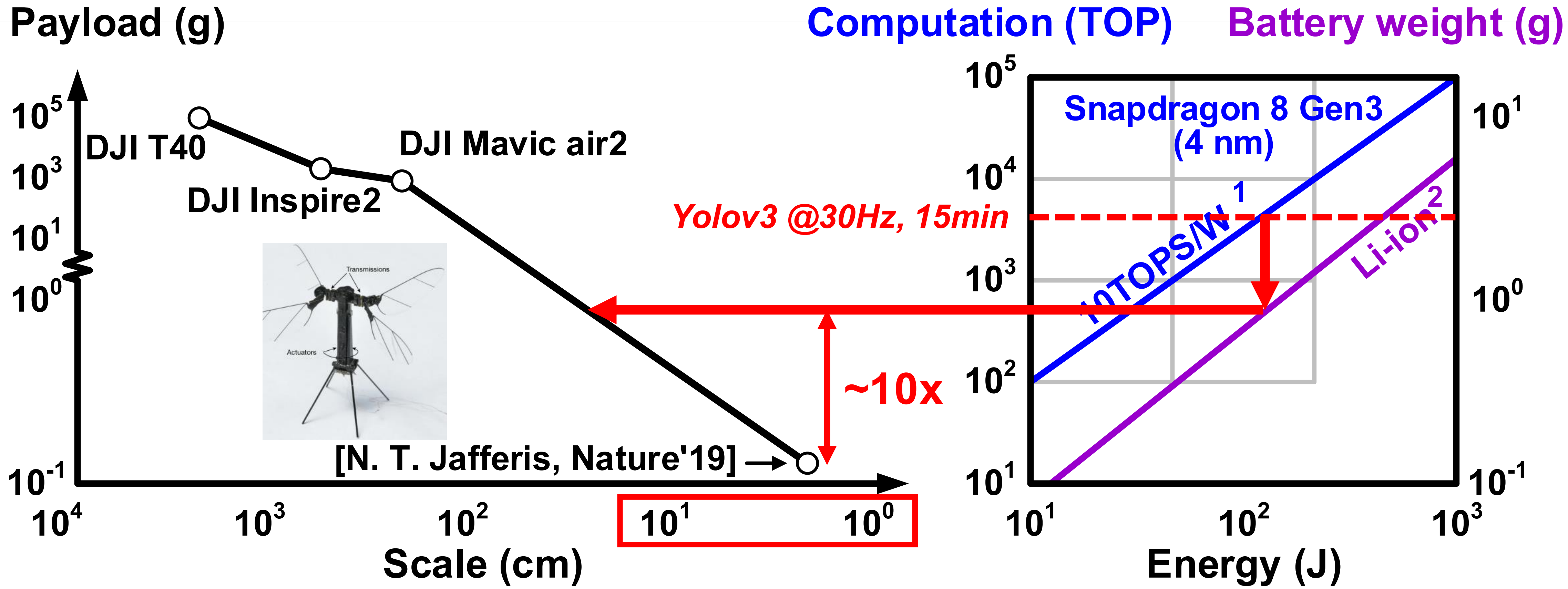




- **Typical autonomous system in robots**



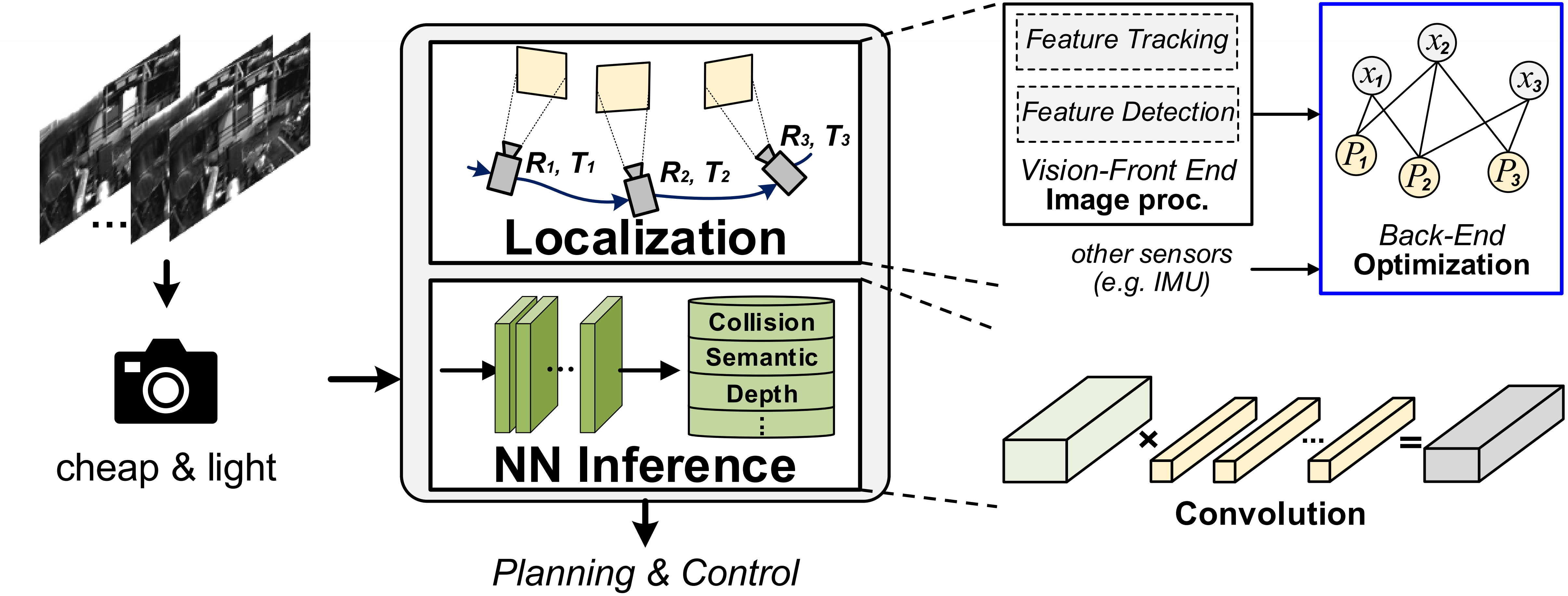
Miniaturization vs. intelligence



[1] From https://nicsefc.ee.tsinghua.edu.cn/projects/neural-network-accelerator.html. [2] From *[J. B. Quinn, JES'18]* w/ 50% energy for computation

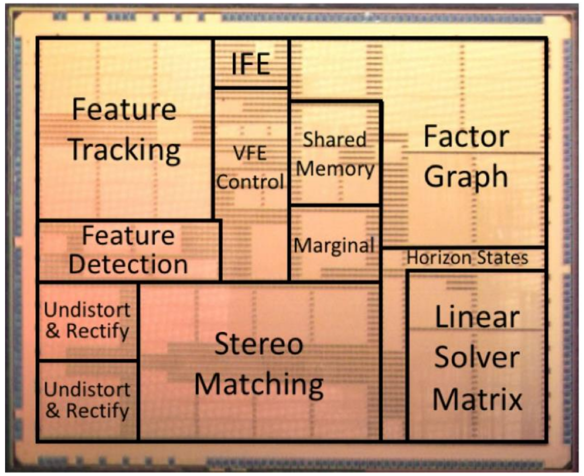
Require highly energy/area-efficient processers

Visual perception in microrobots

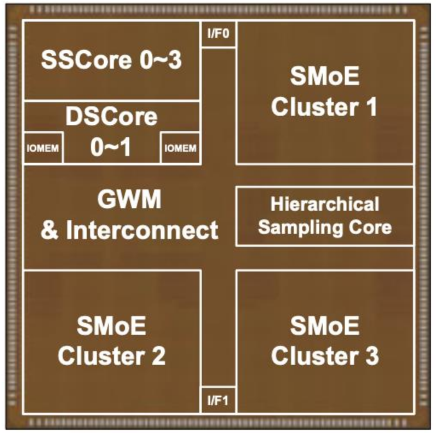


Introduces significant computation: NN inference & Localization

Pioneer Microrobotic VP Processors

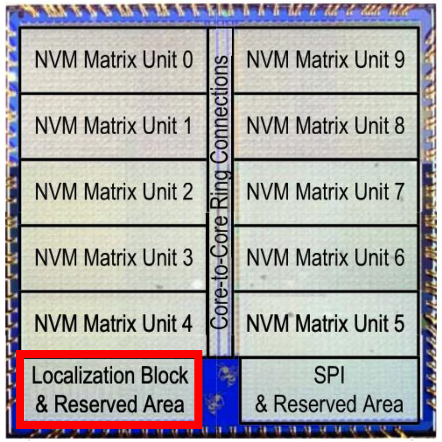


[A. Suleiman, JSSC'19]
Pioneer work on energy-efficient visual-inertial odometry w/ specific data sparsity



[G. Park, ISSCC'24]

ASIC design for advanced NN-based localization algorithm to achieve higher accuracy



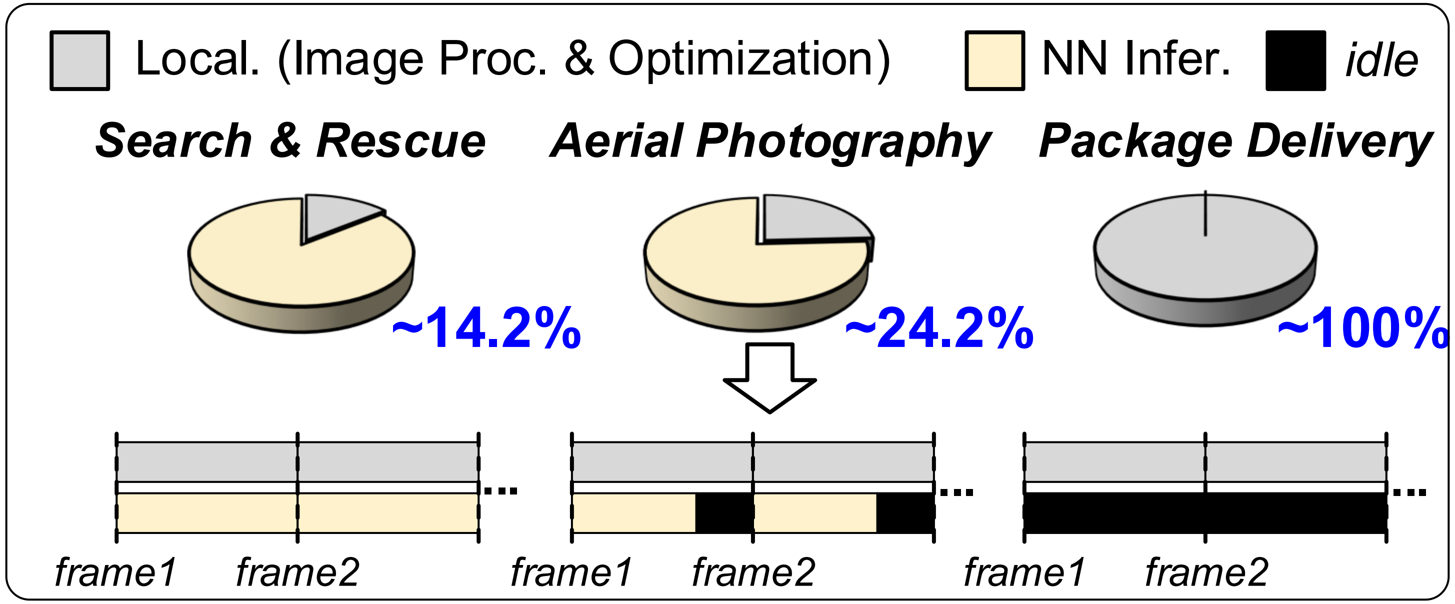
[S. D. Spetalnick, ISSCC'24]

ASIC design for bristle robots' surveillance w/ matrix units and an independent localization block

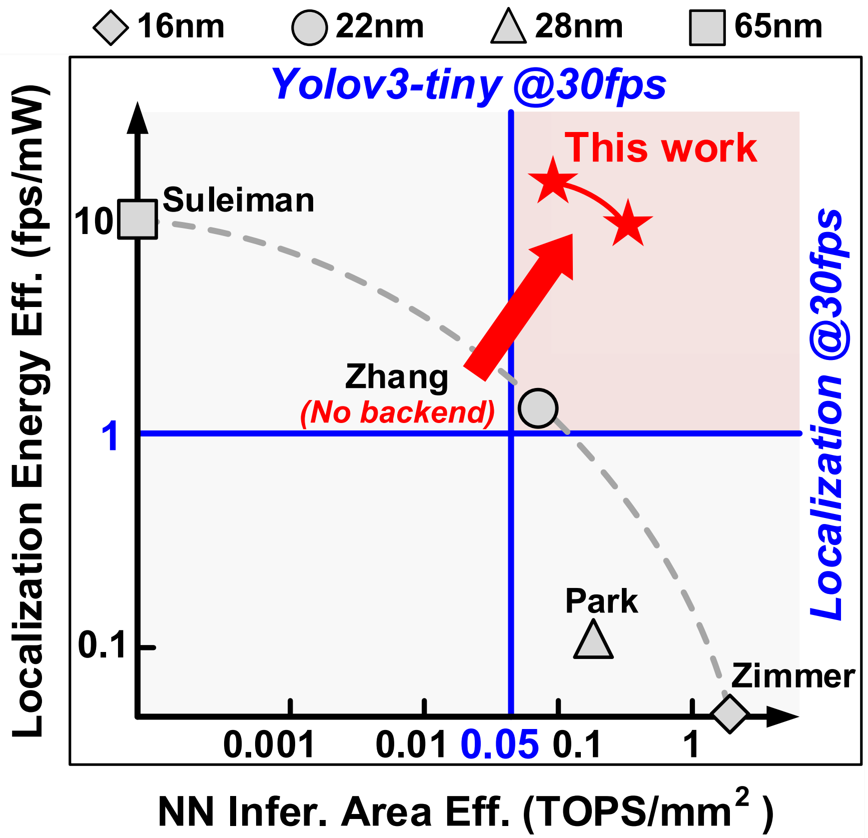
Prior VP processor (VPP) works: specific algorithm or separate hardware

Limit. of separate hardware

Computation ratio & thru. degradation in diverse tasks



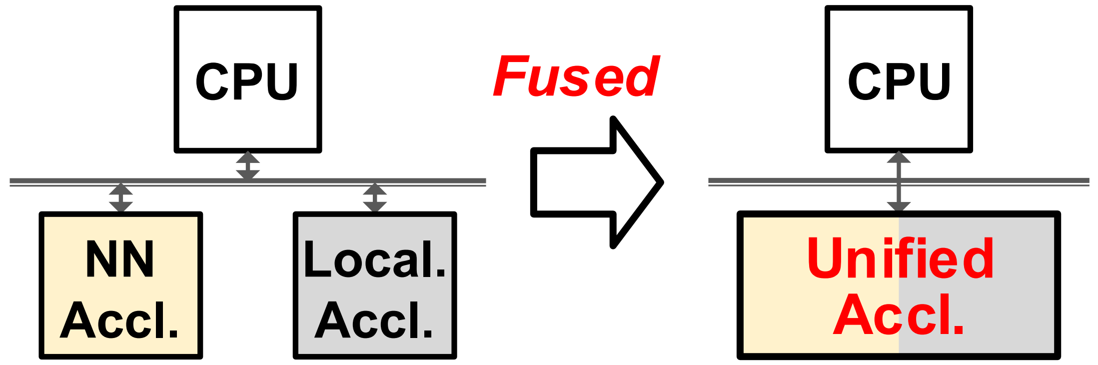
Data from **[B. Boroujerdian, TOCS'22]**



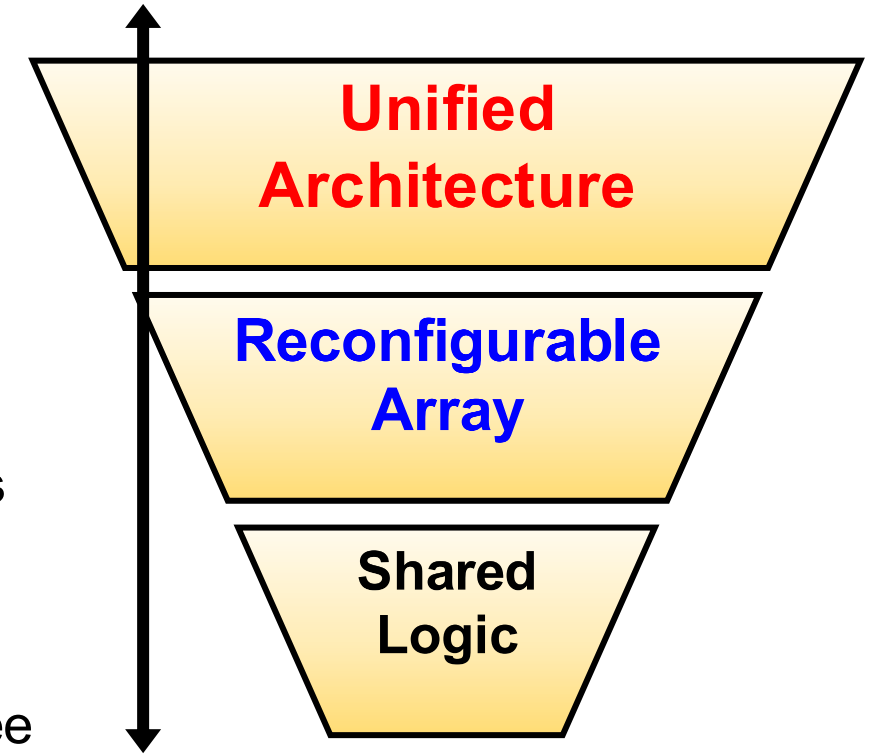
VPP should support flexible function switching

Motivation: Unified VPP

Merging two separate computation nodes into one



Design Hierachy



Target: system-level performance in diverse robotics tasks

- Architecture level: VLIW-style ISA & unified dataflow
- Array level: reconfigurable PE array
- Logic level: rearranged buffers & logic-shared adder tree

This work features an architecture-to-logic hardware fusion for unified VPP

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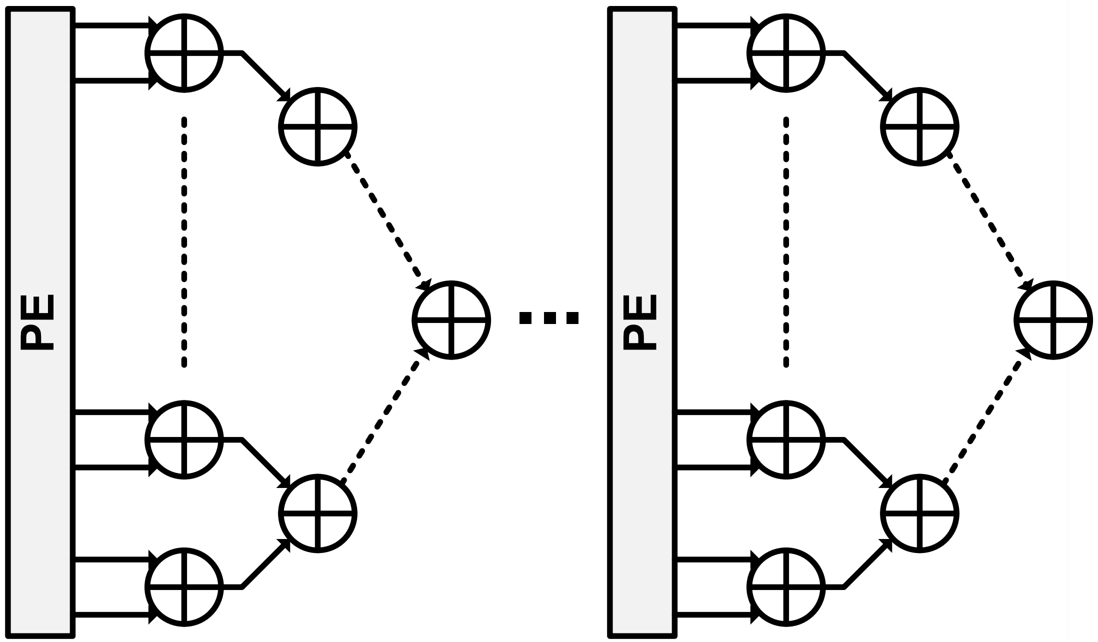
Challenge 1: operator heterogeneity

Matrix-vector-multiplication (MVM)

$$\begin{bmatrix} y_1 \\ \vdots \\ y_M \end{bmatrix} = \begin{bmatrix} w_{1,1} & \cdots & w_{1,N} \\ \vdots & \ddots & \vdots \\ w_{M,1} & \cdots & w_{M,N} \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_M \end{bmatrix}$$

Adder-tree array

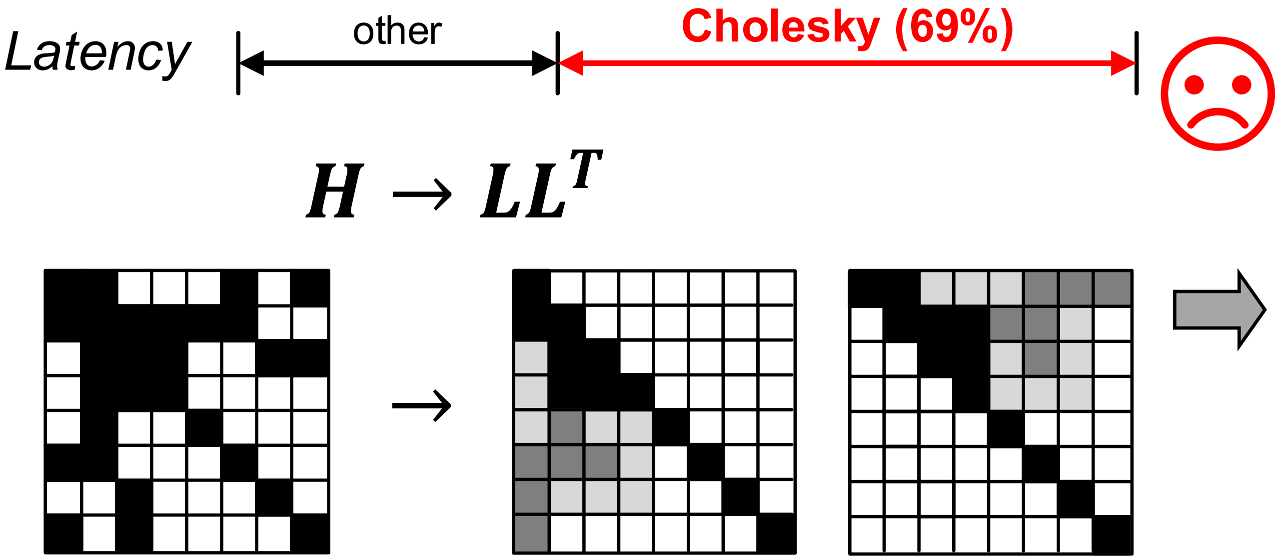


INT8 / BF16

NN Inference can be accelerated by adder tree

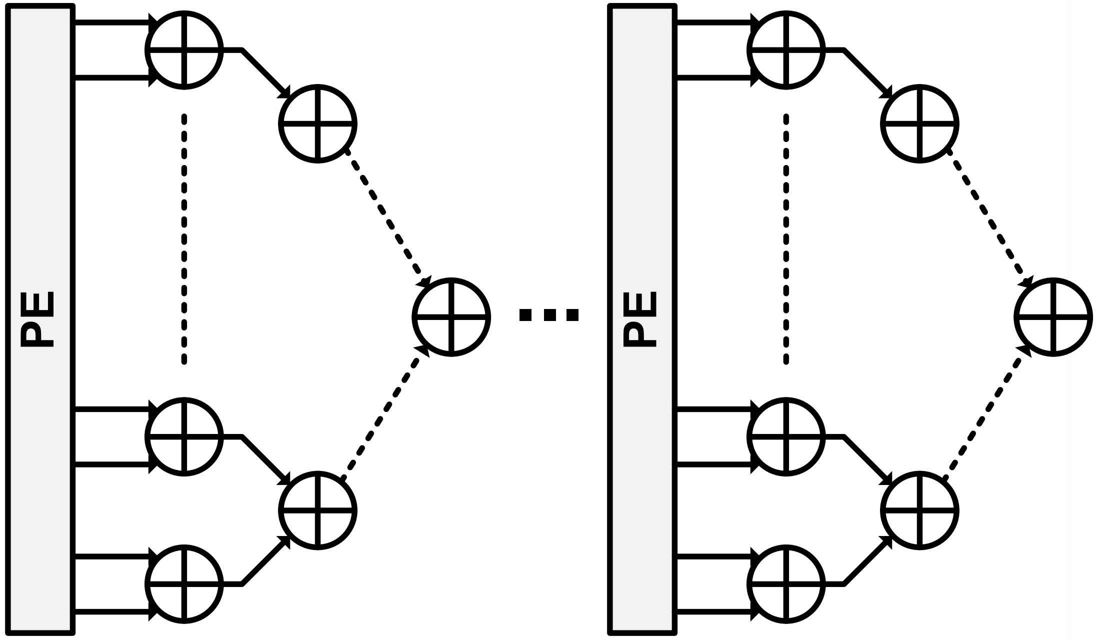
Challenge 1: operator heterogeneity

Cholesky decomp. in localization back-end



FP32 / Double

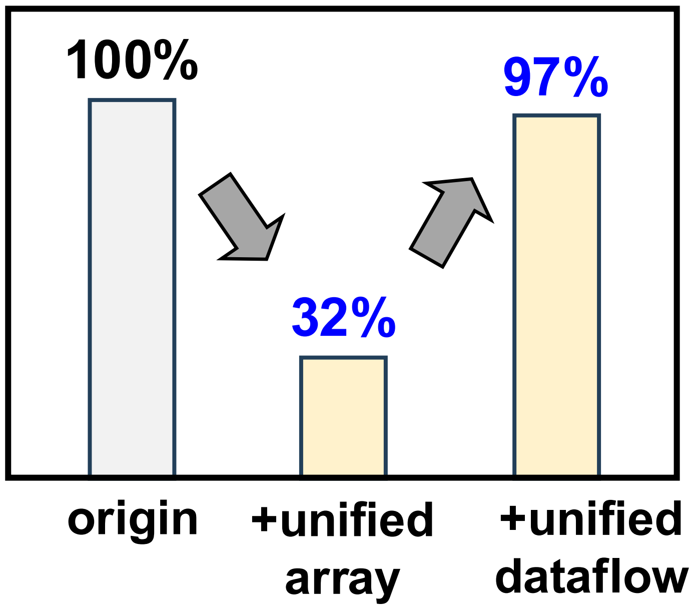
Adder-tree array

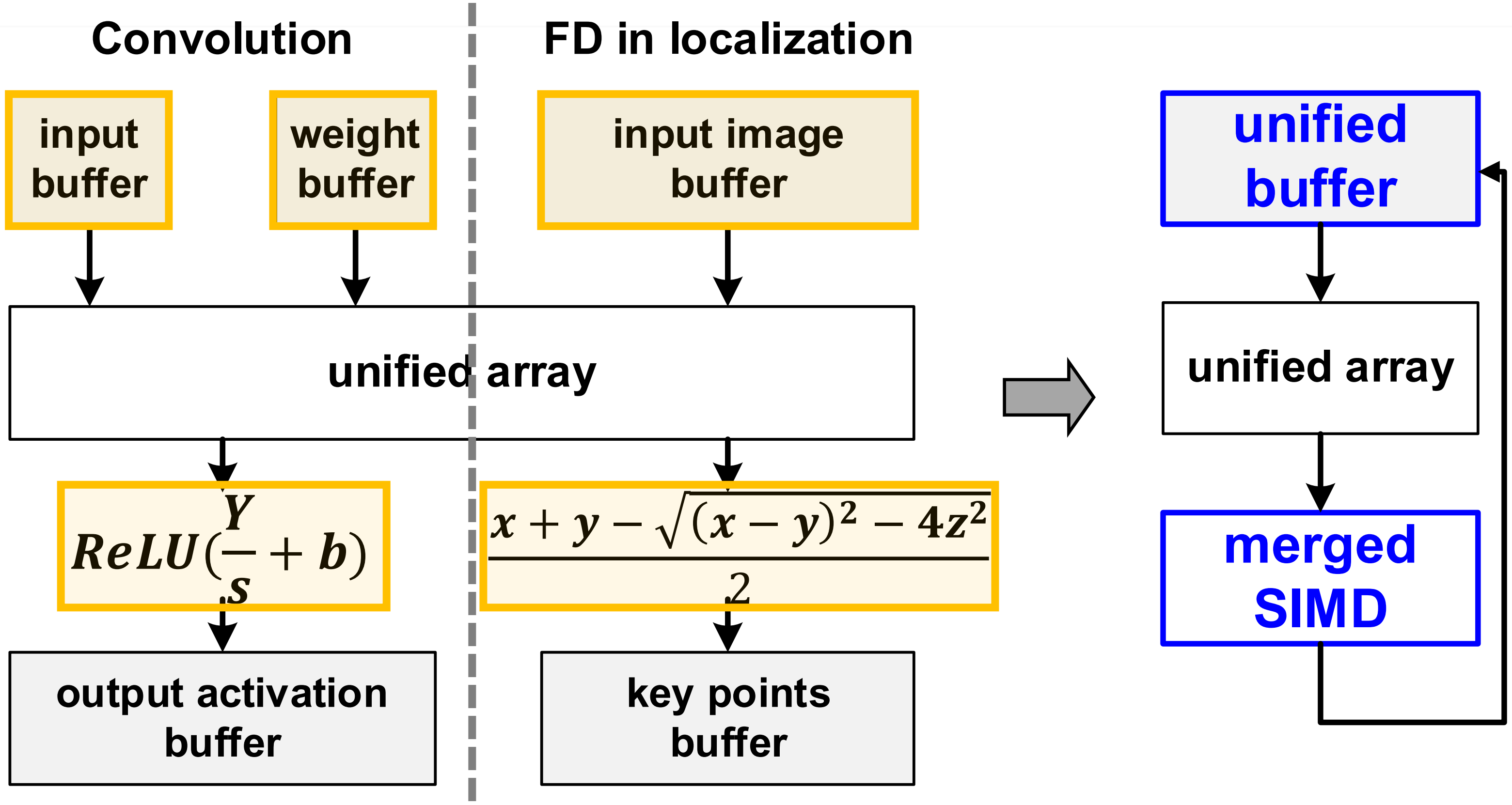


INT8 / BF16

Solution: reconfigurable PE array & triangular-stationary scheme

Challenge 2: dataflow heterogeneity

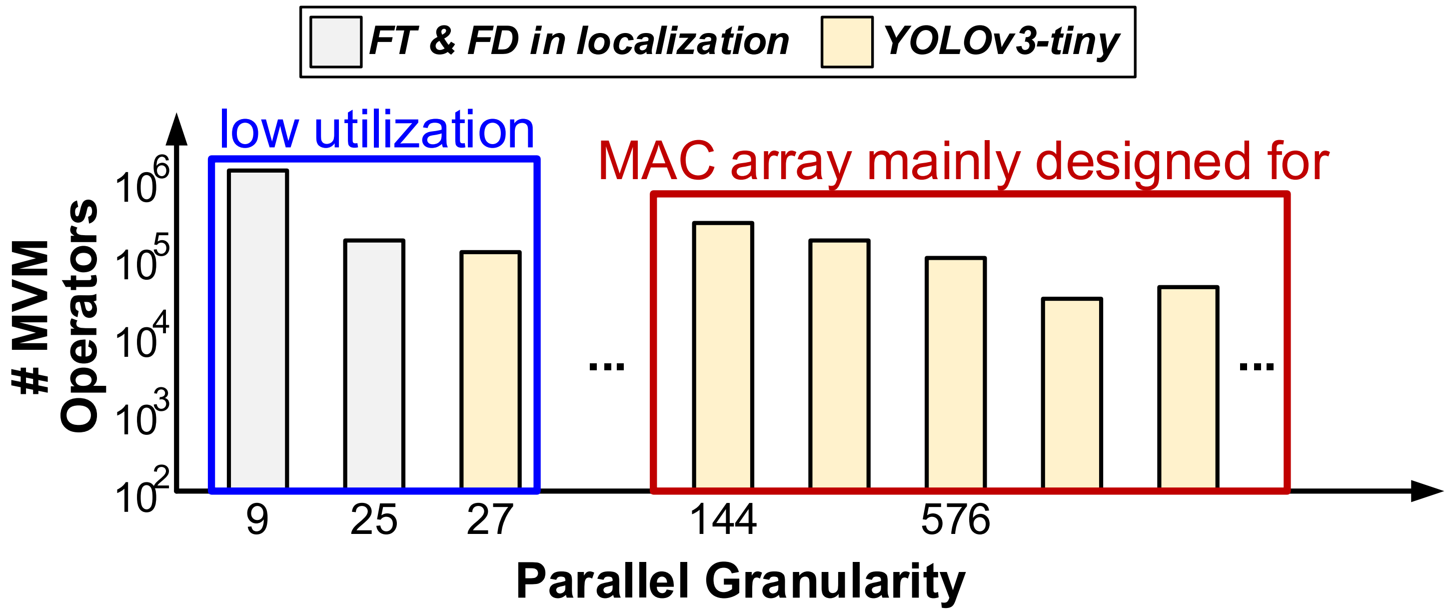




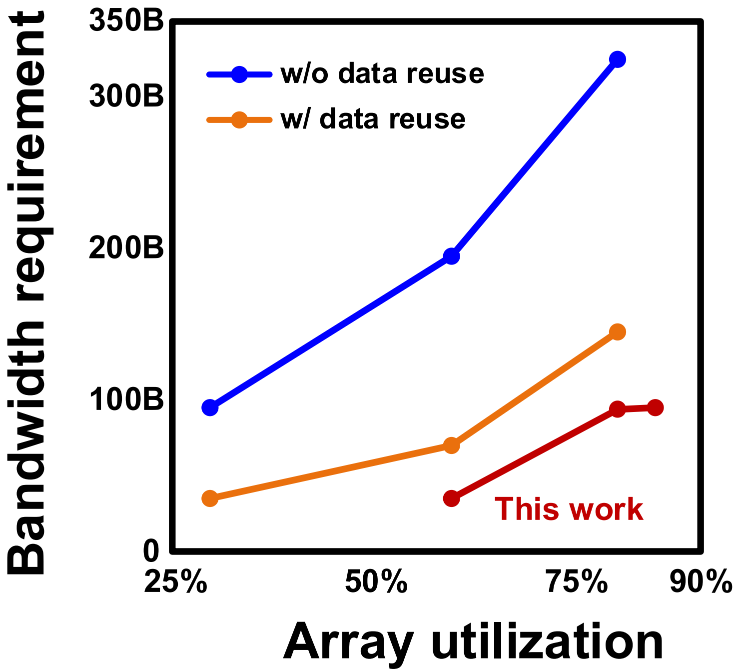
Solution: VLIW-style ISA & merged post-processing datapath

Challenge 3: diverse granularity

- **Drastically different parallelisms for MVM-based operators**
- **High utilization requires high memory bandwidth**



Estimated on 32x32 MAC array



Solution: flexible data mapping with a multi-granularity operand buffer

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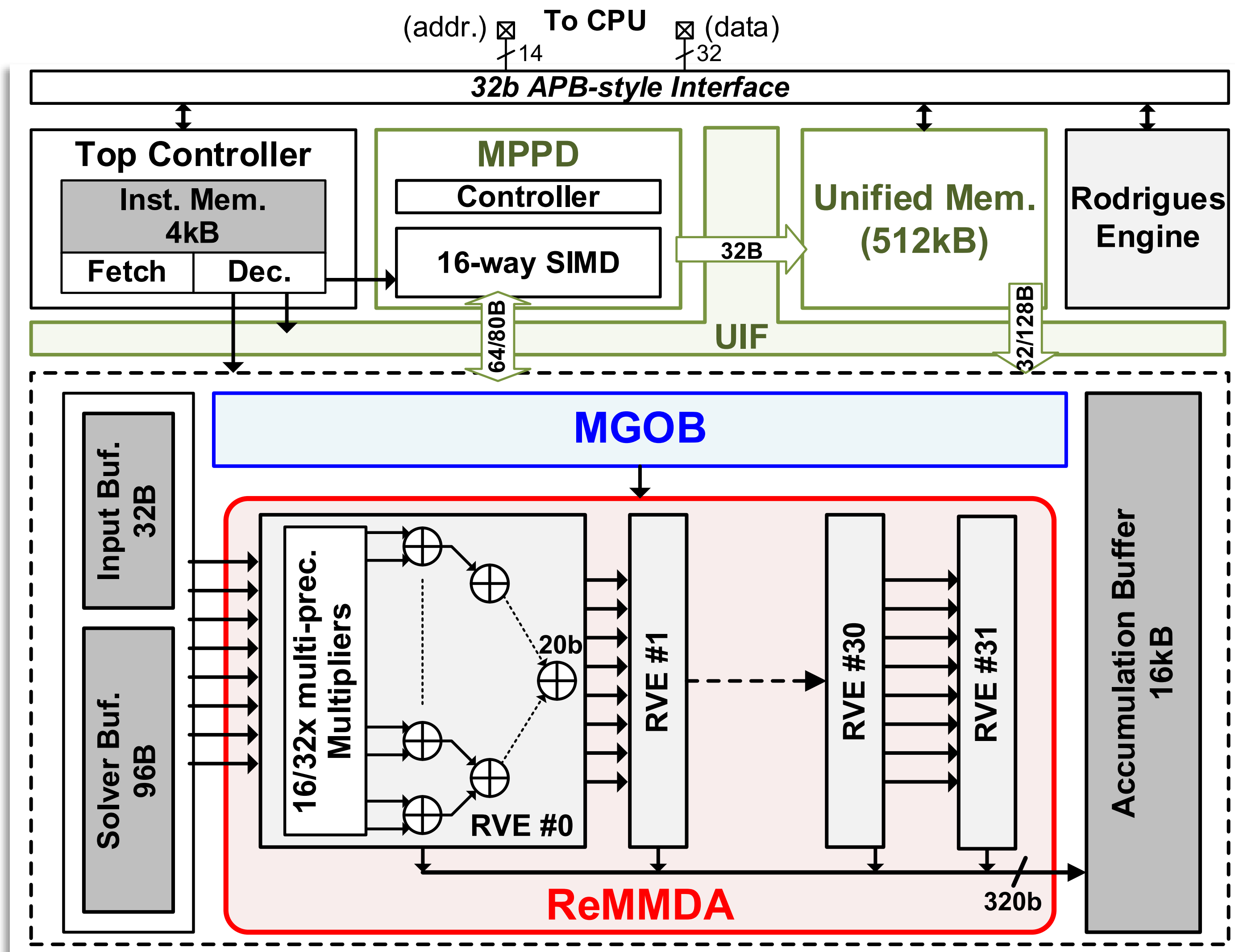
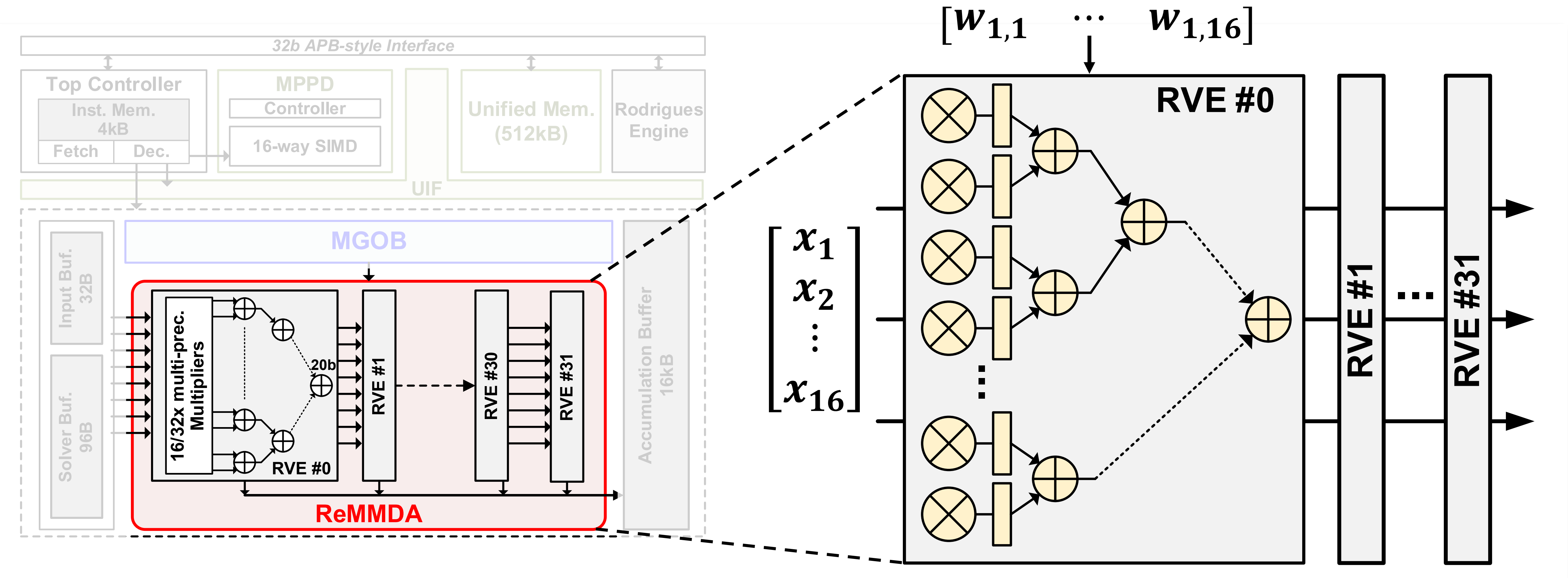
Architecture overview



- A reconfigurable matrix-multiply-decompose array (ReMMDA) for fast Cholesky decomposition with 32x reconfigurable vector engines (RVEs)
- A unified interconnect fabrics (UIF) reduces local SRAM with a merged post-processing datapath (MPPD) & a VLIW-style ISA
- A multi-granularity operand buffer (MGOB) performs flexible data reuse

ReMMDA for MVMs

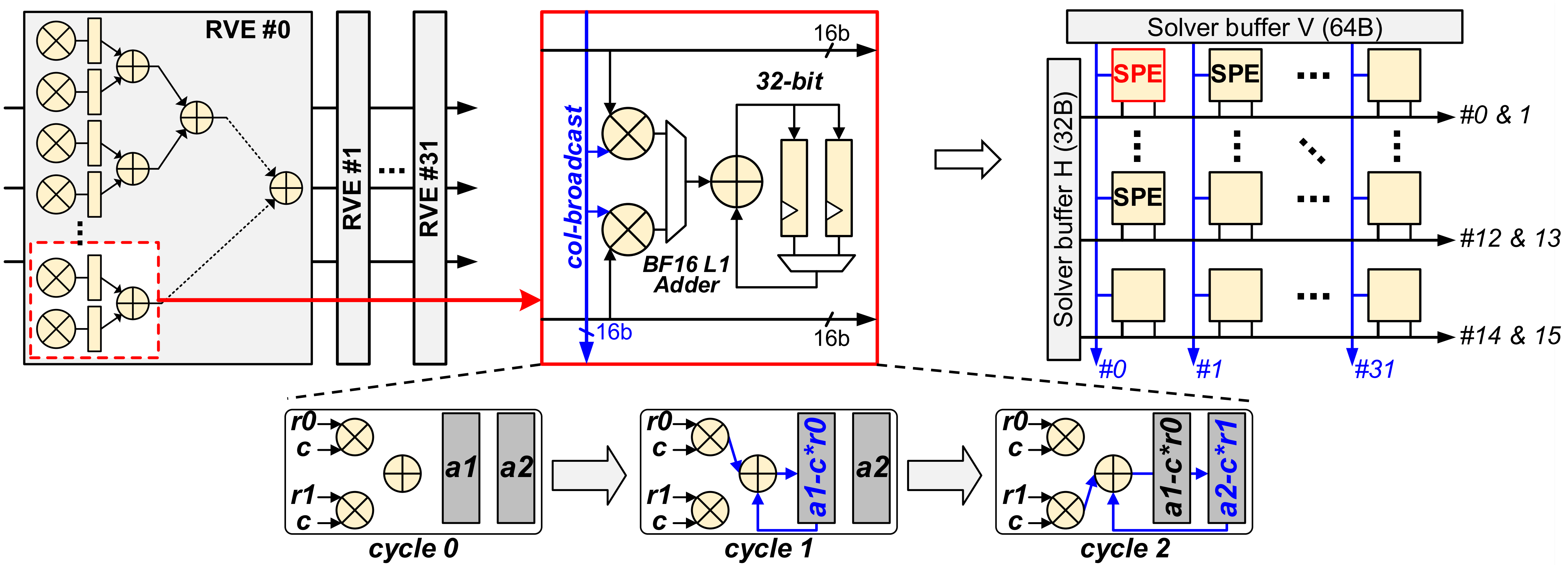


ReMMDA is inherently configured as a 16x32 MAC array in BF16

SPE for fast Cholesky decomp.

Solver-processing element (SPE)

8x32 SPE array



SPE can perform in-situ updating w/ 2D-broadcast & stage registers

Triangular-stationary scheme

$$H = \begin{bmatrix} l_{11}^2 & h_{12} = h_{21} & \cdots & h_{1n} = h_{n1} \\ l_{11}l_{21} & l_{21}l_{21} + l_{22}^2 & \cdots & h_{2n} = h_{n2} \\ \vdots & \vdots & \ddots & \vdots \\ l_{11}l_{n1} & l_{n1}l_{21} + l_{22}l_{n2} & \cdots & \sum l_{ni}^2 \end{bmatrix}$$

$$\Downarrow$$

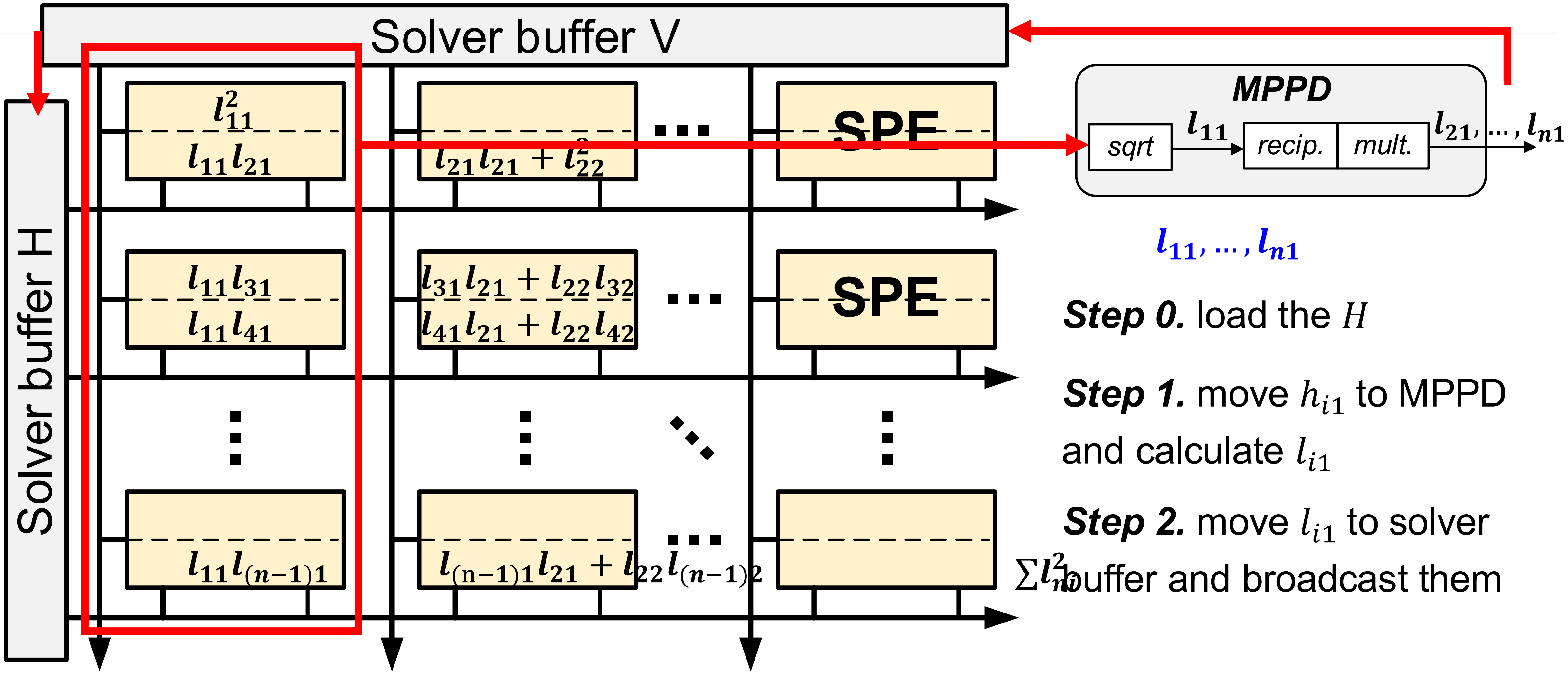
ReMMDA

$$\Downarrow$$

$$L = \begin{bmatrix} l_{11} & 0 & \cdots & 0 \\ l_{21} & l_{22} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ l_{n1} & l_{n2} & \cdots & l_{nn} \end{bmatrix}$$

Step 0. load the H

Triangular-stationary scheme



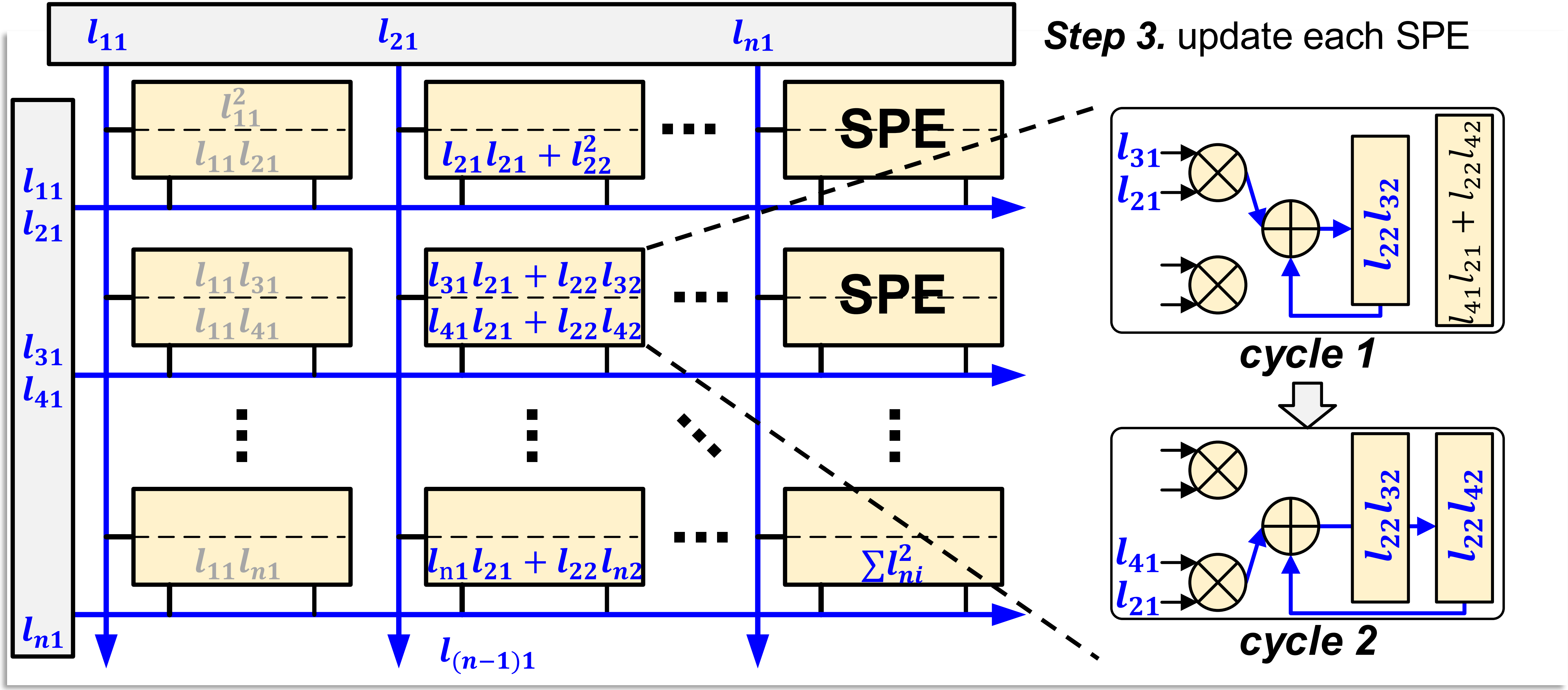
Step 0. load the H

Step 1. move h_{i1} to MPPD and calculate l_{i1}

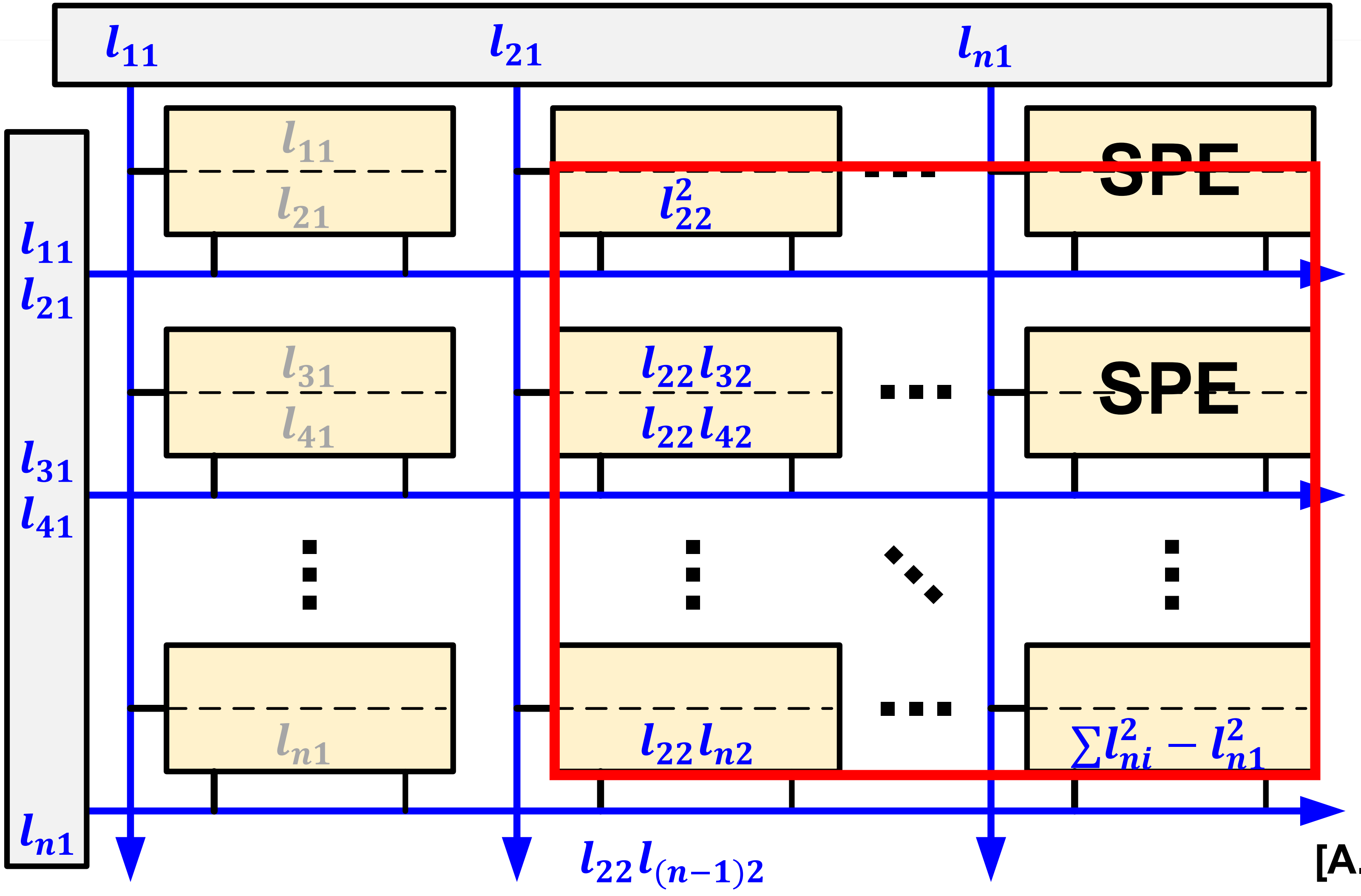
Step 2. move l_{i1} to solver buffer and broadcast them

Triangular-stationary scheme

Step 3. update each SPE

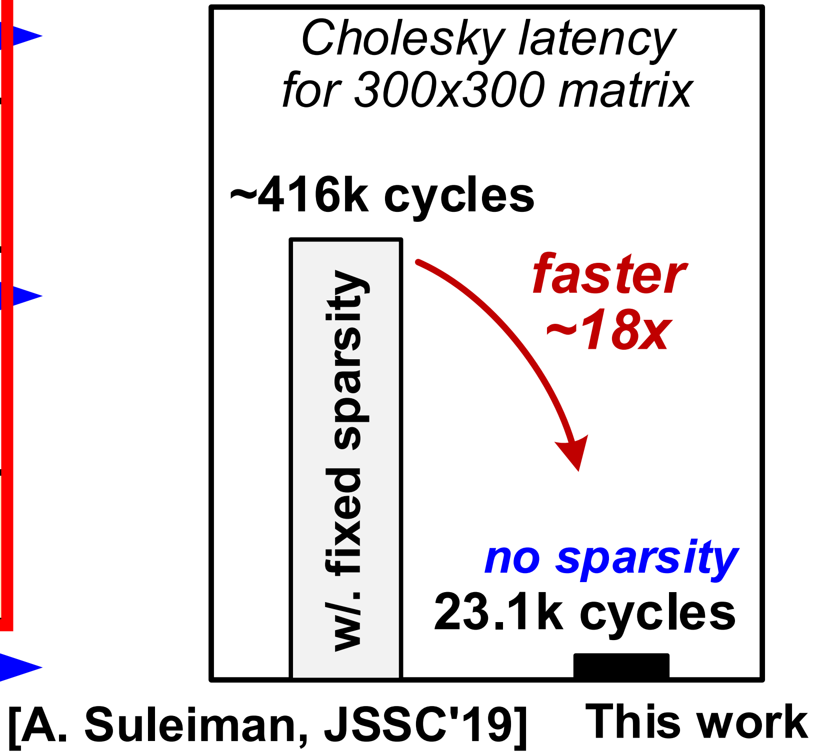


Triangular-stationary scheme

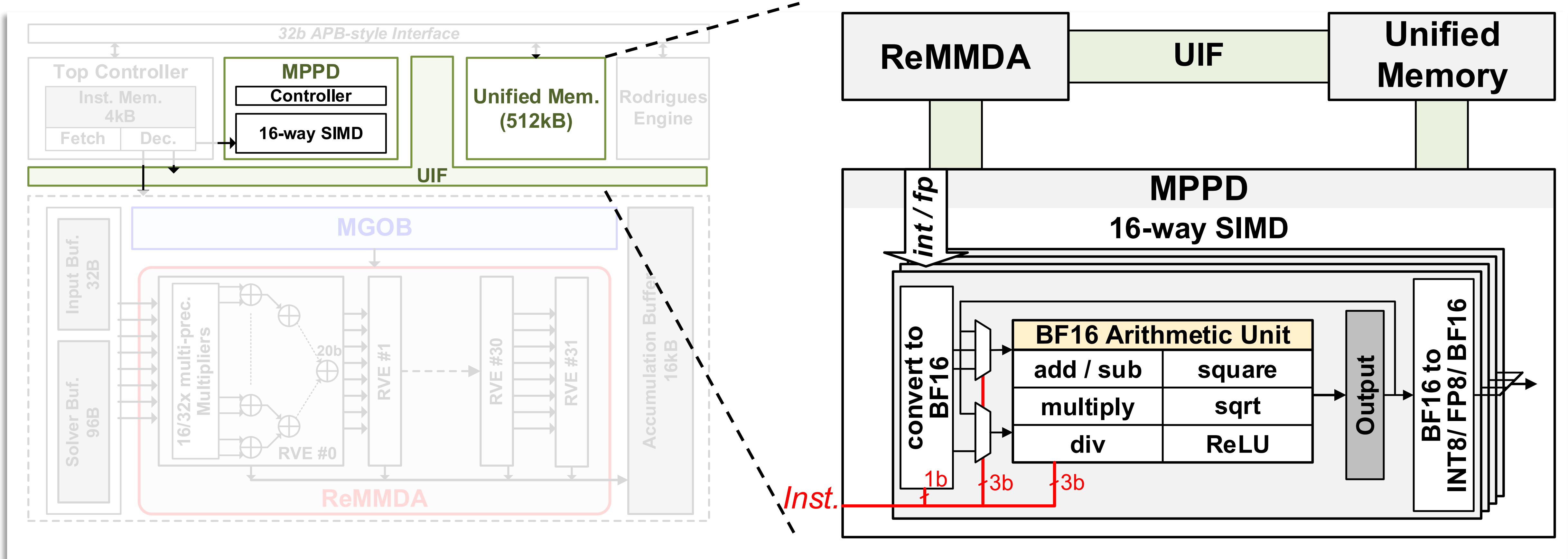


Step 3. update each SPE

Step 4. return to ***Step 1***



Unified dataflow with MPPD



The programmable MPPD converges the remaining computational paths

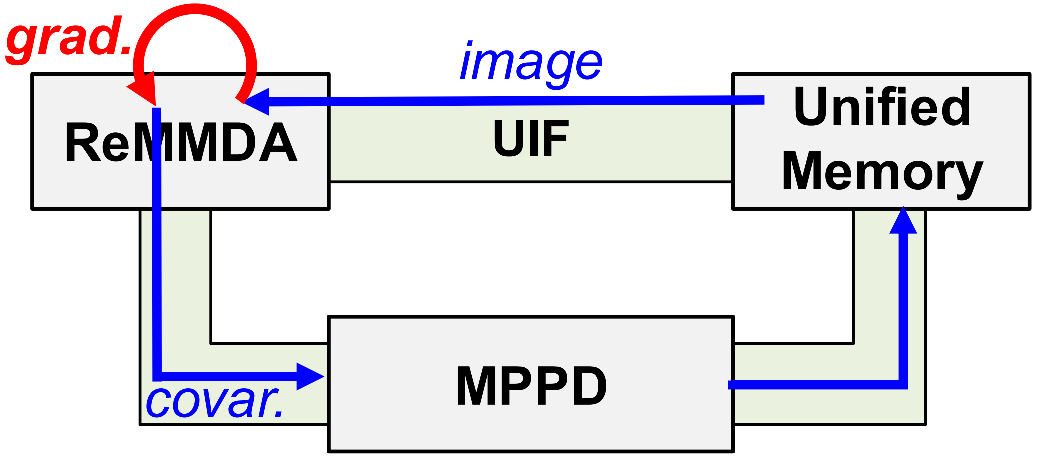
ISA for flexible task mapping

Fetch & Decode → Load Data → Matrix Operation → Post-Processing → Write Back

32b VLIW-style Instruction Breakdown

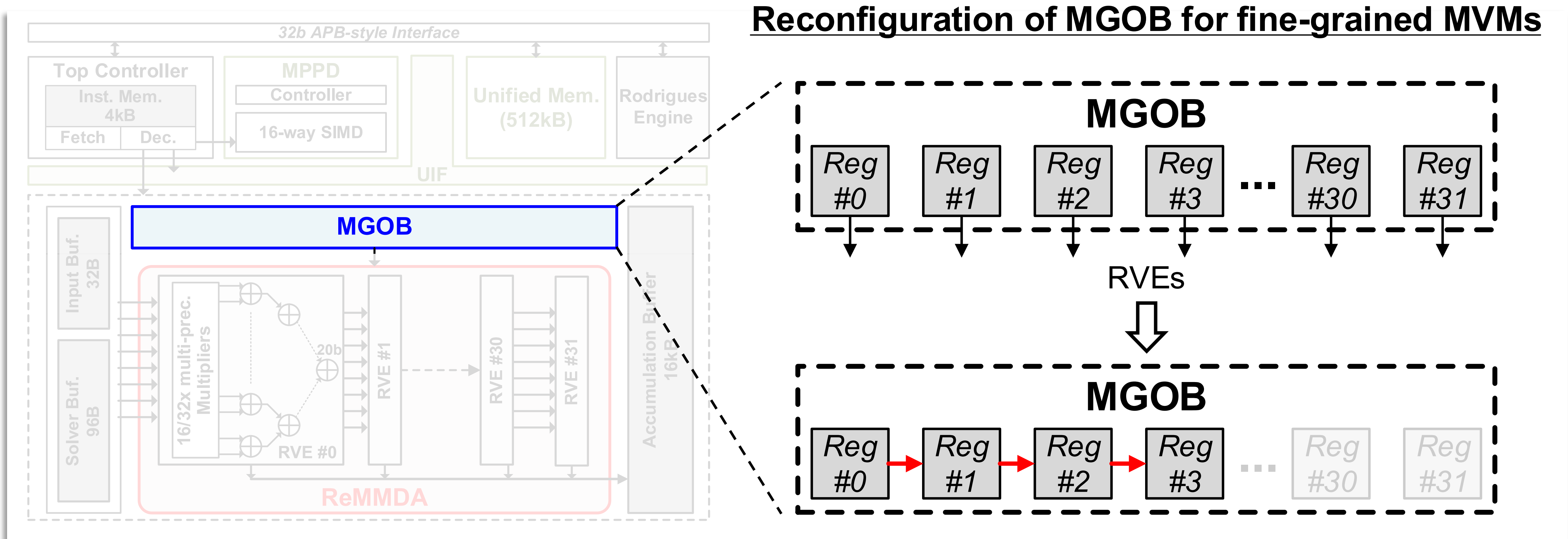
Segment.	31: 24	23: 18	17: 9	8: 4	3: 0
Part	Control	ReMMDA	MPPD	Unified Mem.	Accu. Buf.
Opcode	1b	3b	7b	3b	2b
&Src.RF		1b	1b	1b	1b
&Dest.RF		2b	1b	1b	1b
Imm.	7b				

Example: feature detection



Both NN infer. & local. can be executed by a unified 5-stage computation flow

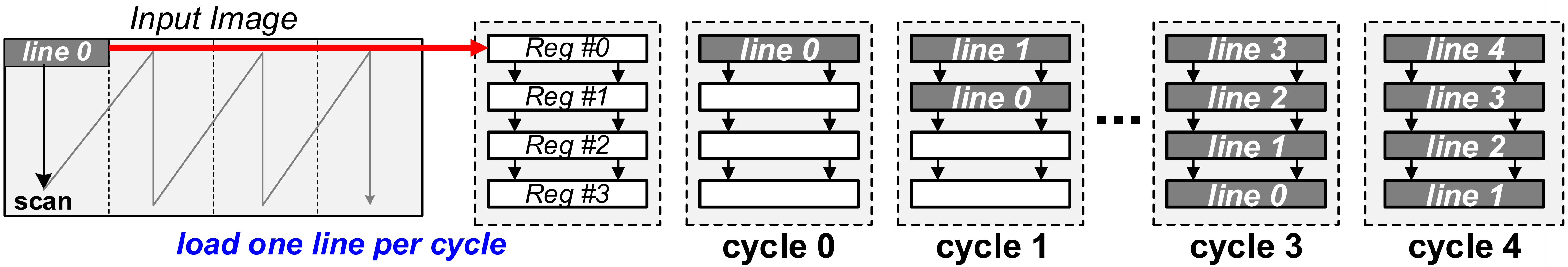
MGOB with flexible data mapping



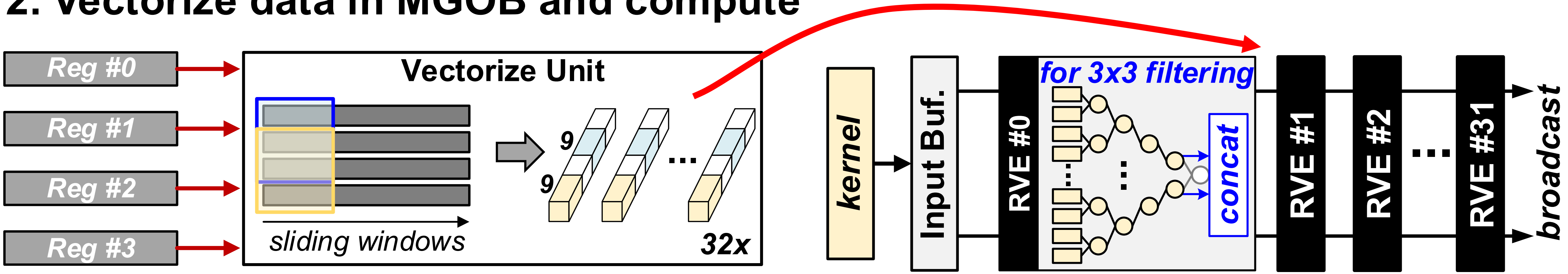
A few register line in MGOB will be arranged as a group of shift-register

Mapping of 3x3 image filtering

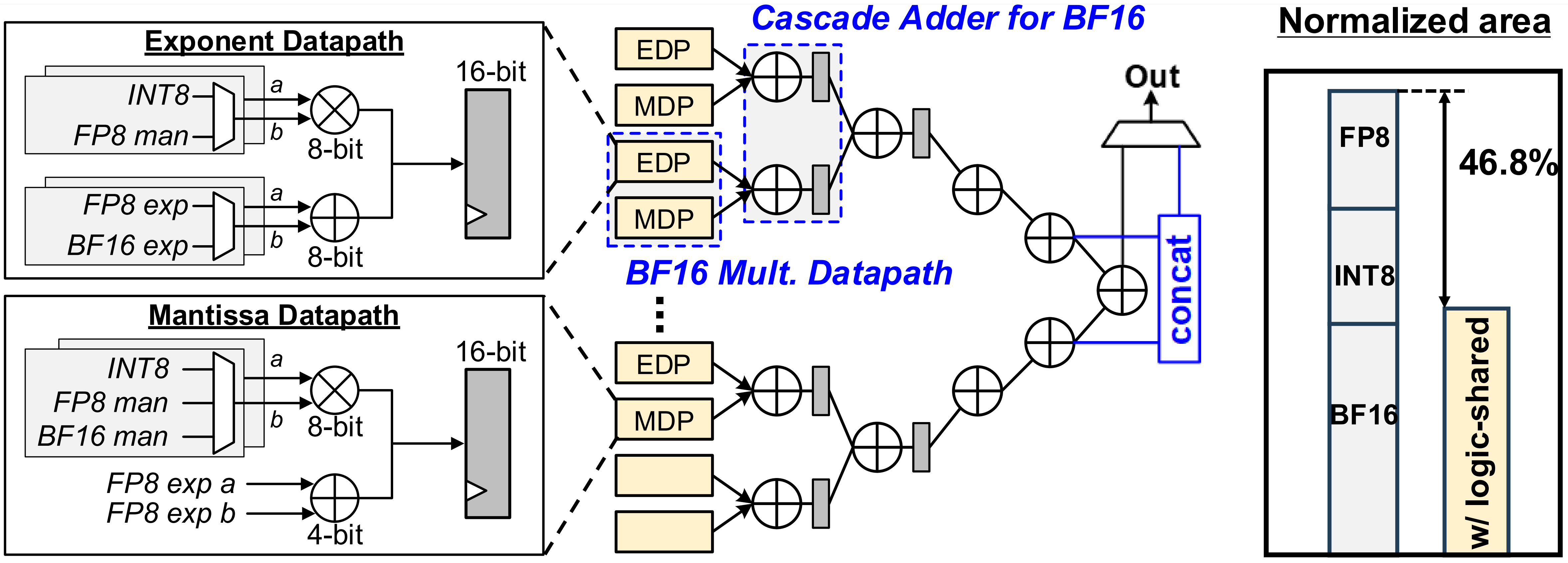
1. Pre-fill the MGOB



2. Vectorize data in MGOB and compute



RVE with logic shared adder tree



Logic sharing reduces the area of RVEs

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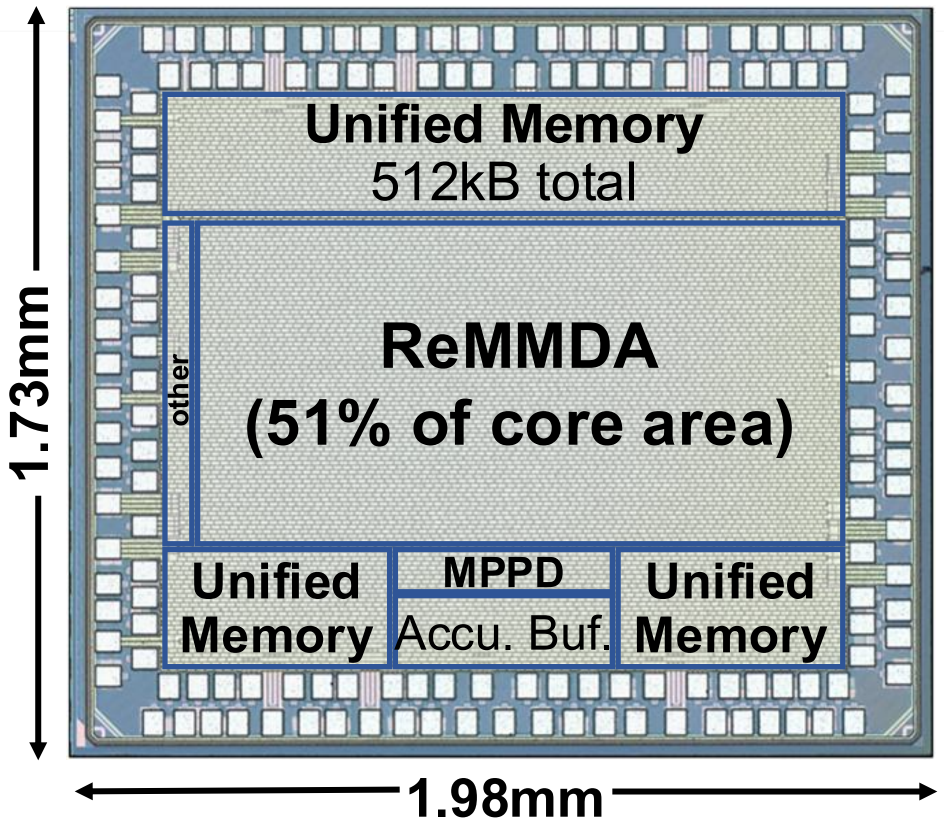
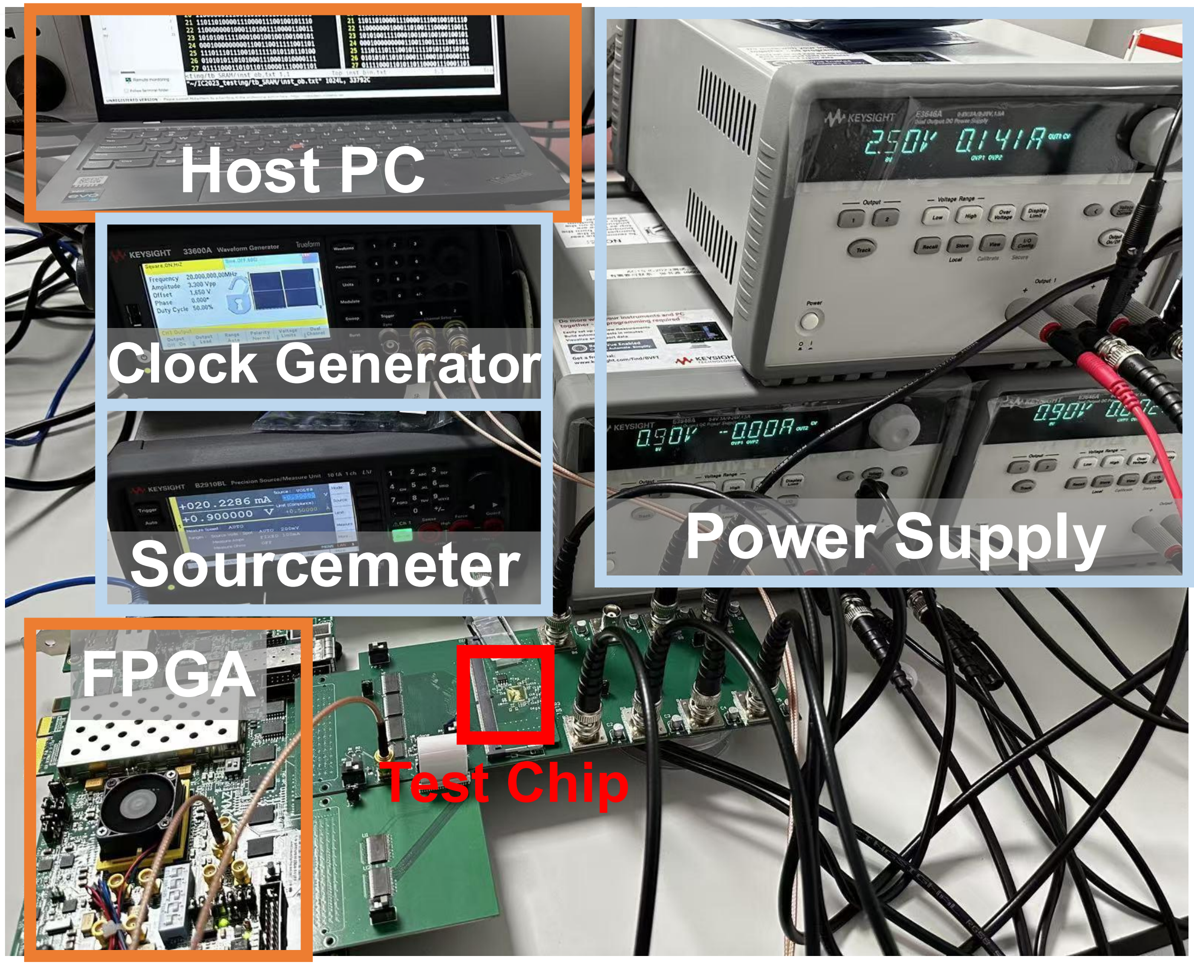
Chip measurements

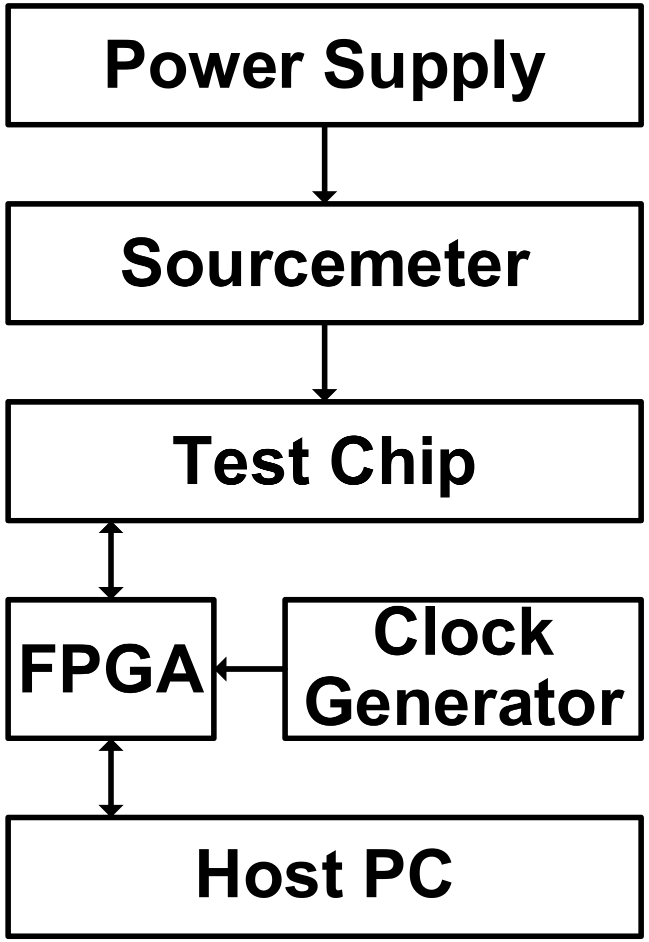


*Includes control operations running on a zero-riscy CPU [P. D. Schiavone, PATOMS'17] @800MHz (1V, ~14.5pJ/cycle) implemented on another 28nm chip w/o parallelism.

Chip Summary	
Technology	28nm
Chip Area	3.425mm^2
On-chip SRAM	532kB
Supply Voltage	0.6-1.2V
Frequency	150-650MHz
Power	24-560mW
Supported Navigation Tasks	Visual localization & NN inference
Best System * Throughput Density	90.9FPS/mm^2
Best System Energy Efficiency	103uJ/frame

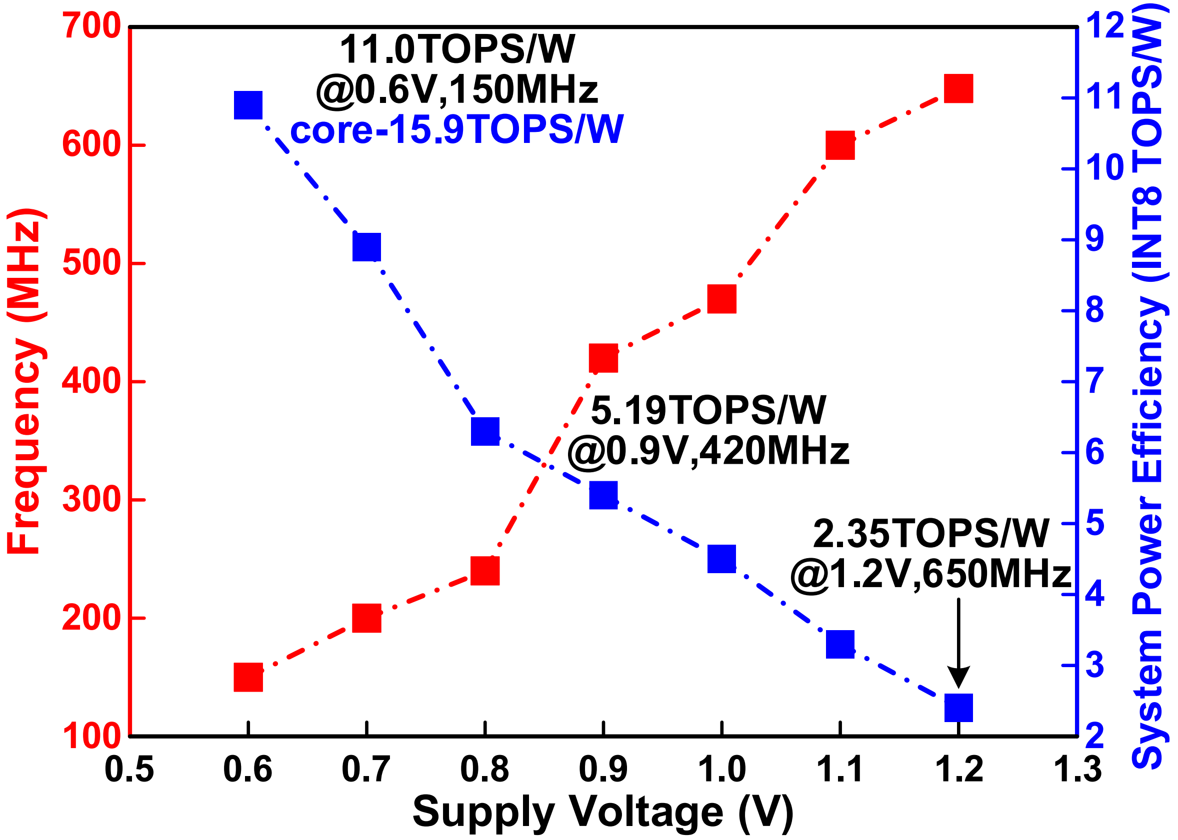
Test platform





NN inference evaluation

Voltage scaling for CONV-BN layer*



Energy breakdown @0.9V, 250MHz (pJ/OP)*

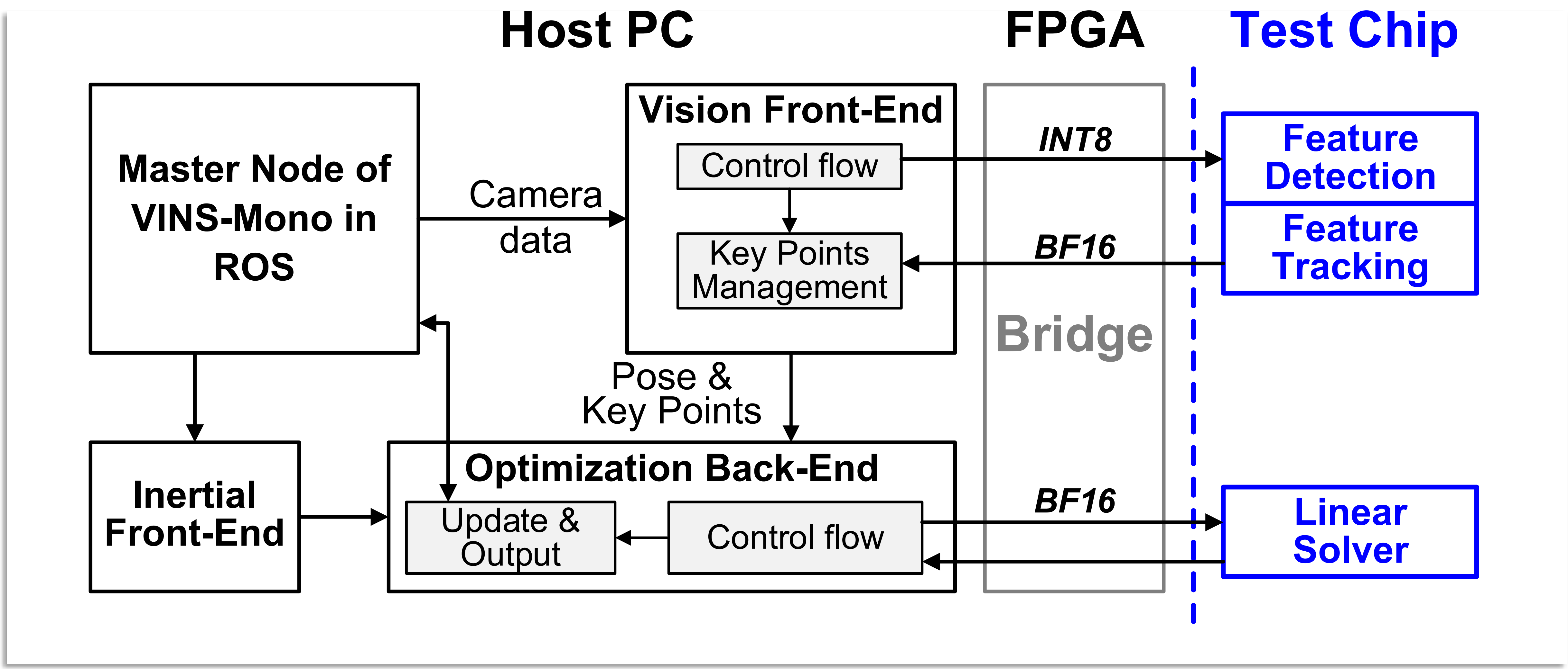
Unified mem.	0.042	MGOB	0.015
ReMMDA	0.062	Accu. Buf.	0.028
MPPD	0.024	other	0.022

Accuracy evaluation results

Model	YOLOv3-tiny	Dataset	COCO2014
mAP50 *(official FP32)*	36.1%	mAP50 *(Ours INT8)*	35.8%
Highest Throughput	219fps, 560mW @1.2V, 650MHz	Best Efficiency	33fps, 18.6mW @0.6V,100MHz

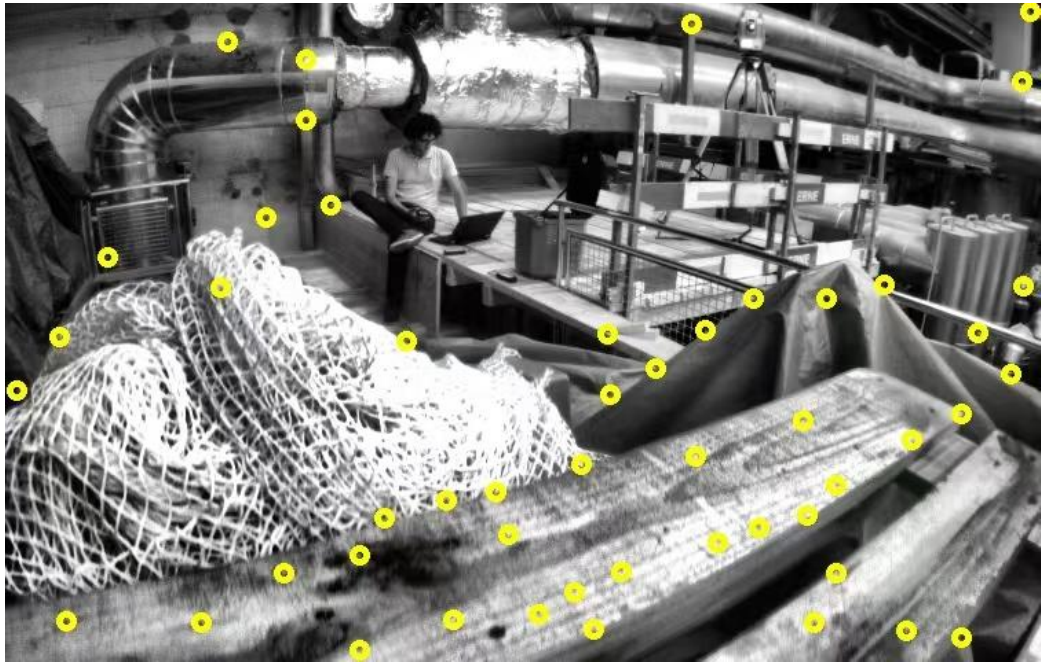
**Measured with 50% weight sparsity & 80% input activation sparsity, and weights are loaded into the array every 128 cycles.*

Evaluation setup for localization

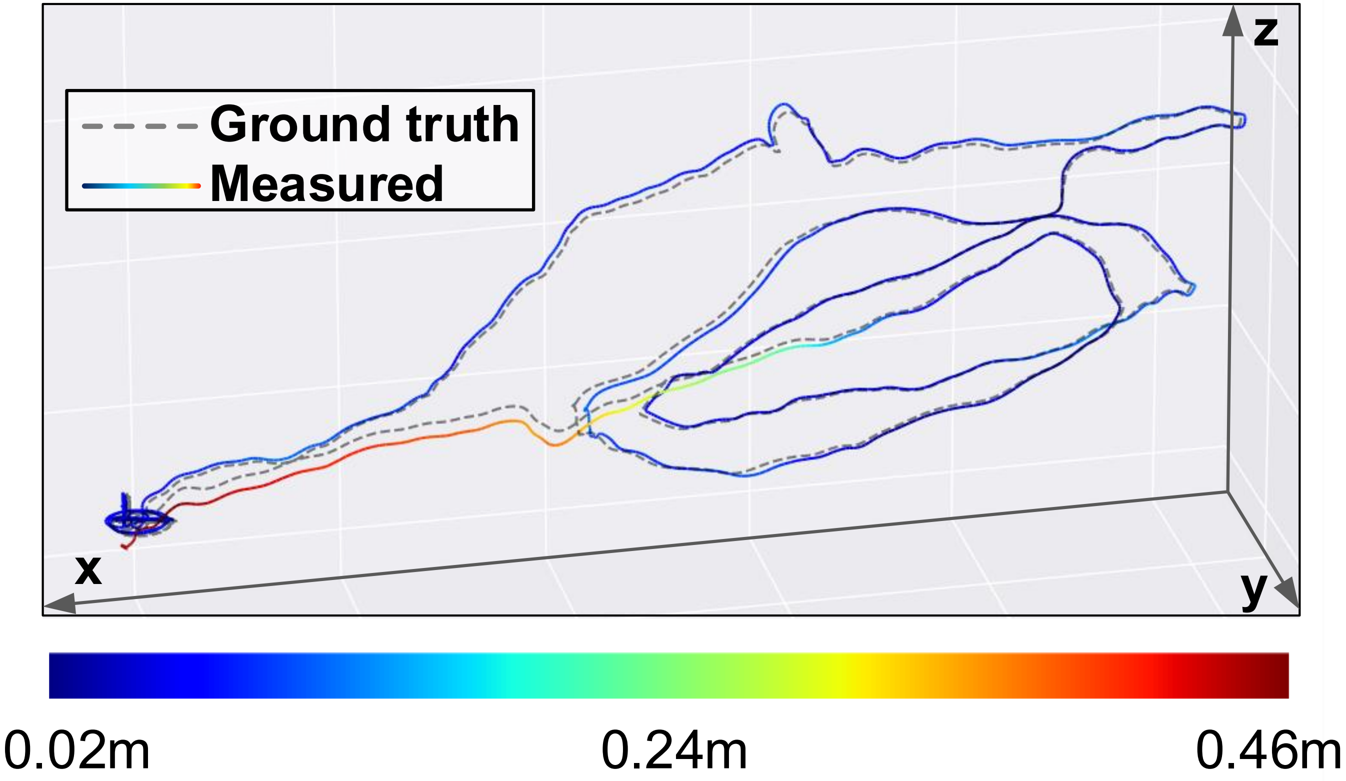


Demo of localization

Feature detection



Trajectory evaluated on EuRoC MH_02



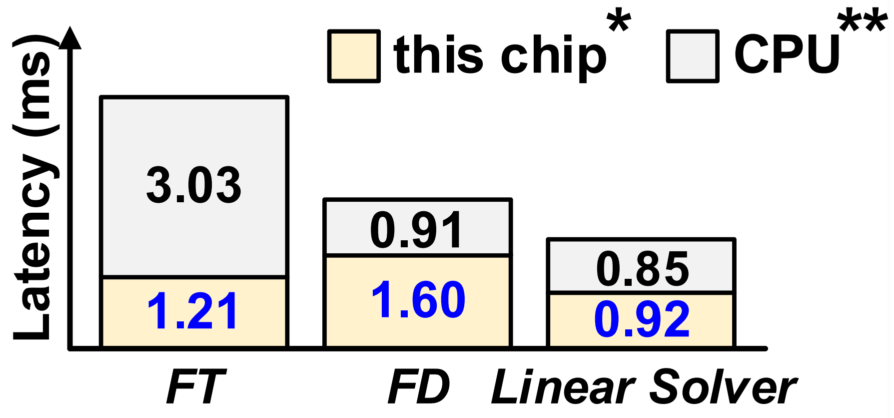
Localization evaluation

Related Trajectory Error on EuRoC

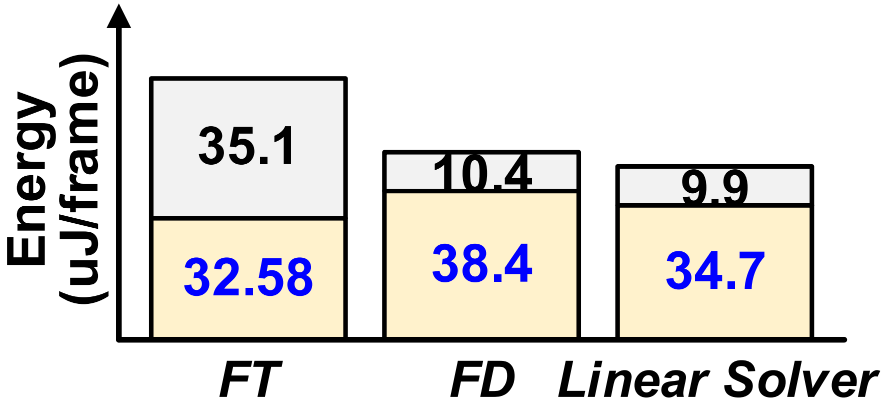
Room #	This Work	Intel i7-12700K
MH_01	0.166%	0.159%
MH_02	0.121%	0.114%
MH_03	0.071%	0.101%
MH_04	0.147%	0.138%
MH_05	0.146%	0.142%
Average	0.147%	0.145%

* @0.6V, 150MHz
**Zero-riscy [P. D. Schiavone, PATOMS'17] implemented on another chip in 28nm (~14.5pJ/cycle, @800MHz, 1.0V) for control flow

Latency Breakdown



Energy Breakdown



Comparison to state-of-the-art

		Suleiman JSSC'19	Zimmer JSSC'20	Zhang VLSI'22	Park ISSCC'24	Spetalnick ISSCC'24	**This Work**
Basic Info	Technology	65nm	16nm	22nm	28nm	40nm	**28nm**
Basic Info	Chip/Core Area (mm^2)	20/16.07	6/3.1	8.76/5.89	20.25/18.49	20.25/18.5	**3.425/2.1**
Basic Info	On-chip Memory	854kB SRAM	640kB SRAM	1428kB SRAM & 2MB MRAM	1348kB SRAM	760kB SRAM & 5MB RRAM	**532kB SRAM**
Basic Info	Frequency (MHz)	62.5/83.3	161-2001	0.059-190	50-200	80-210	**150-650**
Basic Info	Supply Voltage (V)	1.0	0.41-1.2	0.5-1.0	0.7-0.9	0.8-1.1	**0.6-1.2**
Basic Info	Power (mW)	24	30-4160	0.468-158	303.5	0.11-609.3	**24-560**[1]
Local.	Tracking Rate (FPS)	171 (752x480)	Not Supported	217 (640x480)	52.6 (640x480)	-	**235-302**[2] (752x480)
Local.	Lat. of 300x300 Cholesky (ms)	6.7 w/ sparsity	Not Supported	Not Supported	-	-	**0.21-1.36** w/o sparsity
NN Infer.	Data Format	Not Supported	INT8	INT8, INT16	FP16	INT8	**INT8, FP8, BF16**
NN Infer.	Peak Thru. (GOPS)	Not Supported	4.01k	511 INT8	1330	268.8	**307-1331 INT8**
NN Infer.	Area Eff. (GOPS/mm^2)	Not Supported	1.29k	86.7 INT8	71.9	14.58	**146-633 INT8**
NN Infer.	Energy Eff. (TOPS/W)	Not Supported	9.52	12.1 INT8	3.24	0.84	**3.71-15.9**[3] **INT8**

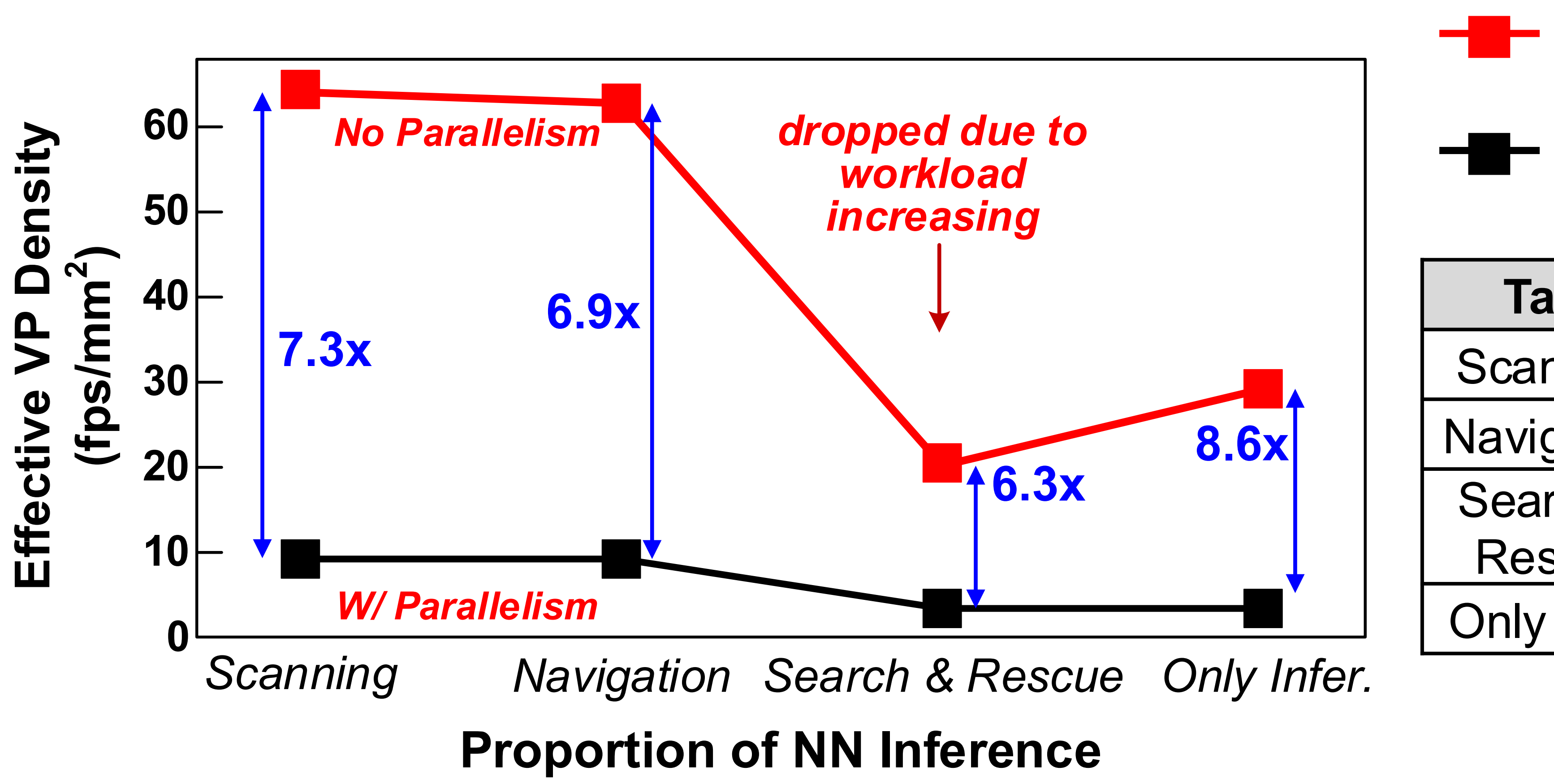
[1] 24mW is measured for feature detection @0.6V, 150MHz, the power of NN inference is 28.7mW @0.6V, 150MHz.

[2] Includes control operations running on a zero-riscy CPU @800MHz (1V, ~14.5pJ/cycle) implemented on another 28nm chip.

[3] With 50% weight sparsity & 80% input activation sparsity @0.6V,100MHz, and weights are loaded into the array every 128 cycles.

System-level evaluation

Effective Visual-Perception Density = Throughput / System Area



Task	Local.	NN Infer.
Scanning	✓	✗
Navigation	✓	Dronet
Search & Rescue	✓	Dronet + YoLov3-tiny
Only infer.	✗	YoLov3-tiny

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Conclusions

- **Energy and area constraints of microrobotic visual perception (VP)**
- **Heterogeneous parts in VP: localization and neural-network inference**
- **Separate hardware suffers from application diversity**
  - Mismatch between dynamic computation ratio and fixed supply
- **Merge two separate accelerators of VP into a unified processor**
  - Array-level fusion: reconfigurability & optimized scheme
  - Architecture-level fusion: unified dataflow & VLIW-style ISA
  - Logic-level fusion: flexible data mapping & logic sharing
- **Unified visual-perception processor in 28nm demonstrates efficient navigation with exceeding system performance**

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